

Name: _____

Exercise 4: Image Classification

In this exercise we'll perform a Supervised Classification of a Landsat TM image. Classification is used to identify and associate spectrally similar pixels in an image dataset. There are two broad approaches to statistically-based classification: supervised and unsupervised.

In a supervised classification you, the analyst, identify groups of pixels that represent areas of known interest: for example, urban areas, water bodies, deciduous forests, and active agricultural fields. You then use one of several possible statistical schemes to place all the other pixels in the image within one of the classes that you have defined. Since the analyst defines the number and spectral character of the classes beforehand, the result of a supervised classification is a dataset of informational classes; classes that are grouped by distinctions of land cover type as judged by the analyst. You may know exactly what a specific class represents, or you may have only a general idea. This can be contrasted with an unsupervised classification, in which the analyst allows the program to define classes on the basis of the statistical distribution of pixel values across the image. When looking at the results of an unsupervised classification, you *may* recognize that certain classes are associated with known features in the landscape, but you do not know this a priori.

For either kind of classification, ground truth is the key to understanding the accuracy of the analysis and the specific nature of the ground cover for each class.

Note that pixels in a classified dataset have arbitrary integer values that represent classes (distinct areas or categories of interest) as opposed to pixels in a raw satellite image that have values proportional to spectral radiance (a measure of radiation sensed by the satellite).

To begin:

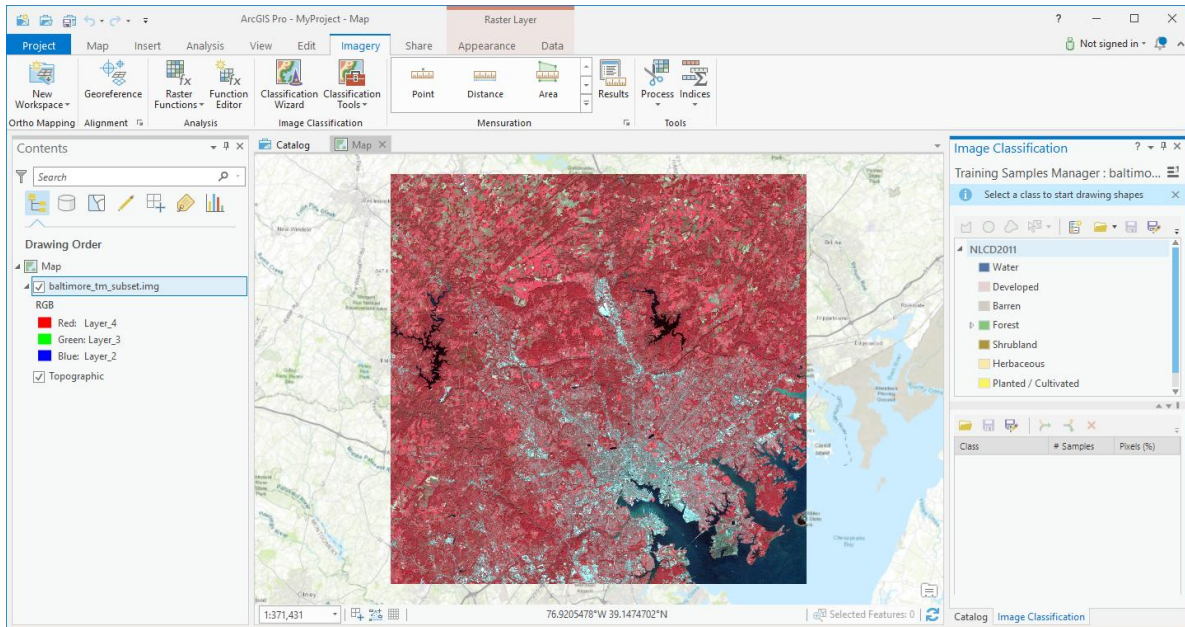
1. Download Exercise4.zip from the link on Canvas and extract all files.
2. Start ArcGIS Pro and create a new Project, using the blank Map template.

Define Training Regions

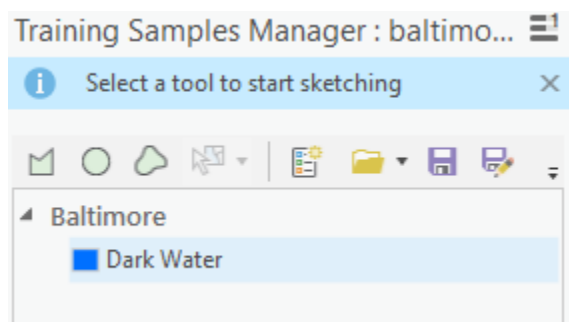
The first step in supervised classification is to define training regions for the various informational classes you hope to distinguish. Training regions are areas of the image that contain "typical" pixels for an informational class. A good training region will include enough pixels to capture the natural variability in the class and will exclude any pixels that should not belong to the class. You will define training regions for eight classes in your Landsat TM scene: **"Dark Water," "Maroon Vegetation," "Pink Fields," "Bright Water," "Dark Green Patches," "Bright Green Patches," "Bright Urban/Industrial," and "Darker Urban/Industrial."** You can see by the names of these classes that we know more about the character and composition of some of the informational classes than others. This is typical and the reason that ground truthing (or effective use of Google Earth) is so critical. *Remember that the goals are to make the*

training regions as large as possible without including any pixels that do not belong in that informational class.

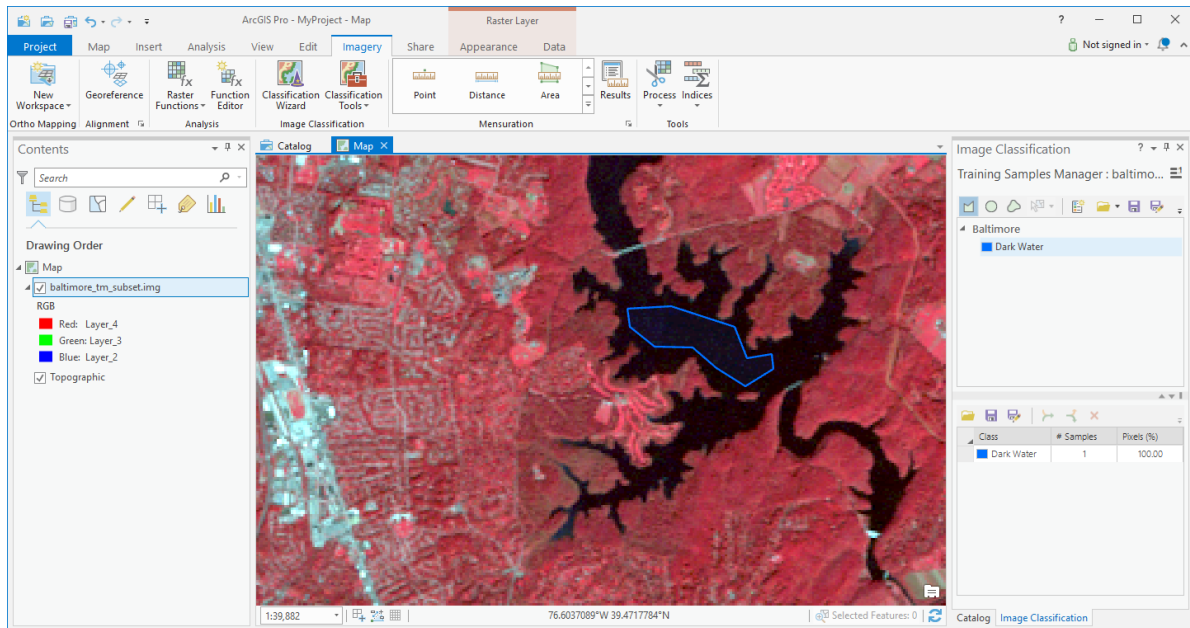
1. Use *Add Data* under the *Map* menu to open the image *Baltimore_tm_subset.img* (saying “yes” to pyramids when prompted), and use *Appearance* → *Symbology* to assign Layer 4 to the red display channel, Layer 3 to green, and Layer 2 to blue: a 432-RGB combination. This Landsat TM image was acquired on October 19, 2009.
2. Open the *Training Samples Manager* under *Imagery* → *Classification Tools*.



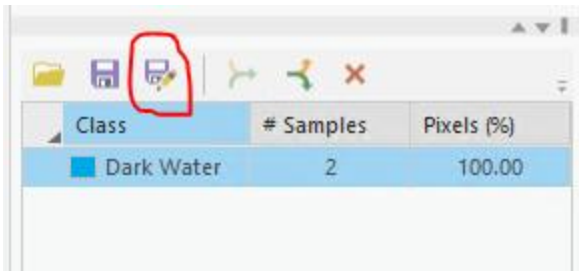
3. By default, the Training Samples Manager will open the National Land Cover Database 2011 (NLCD2011) classification scheme. We want to create our own, so click on the “Create New Schema” icon: 
4. Right click on the words “New Schema” and select “Edit Properties.” Give your new schema a name (e.g., Baltimore2009). Save and close the Properties dialog.
5. Right click on your schema’s name and select “Add New Class.” Name this class “Dark Water,” assign it a value of 1, and assign it a color of your choice. Click “OK” and close the New Class dialog.
6. Click on your “Dark Water” class. This will activate the training region drawing tools.



7. Zoom in to one of the reservoirs in the image. Get close enough that you can clearly distinguish water from shoreline. [Note: holding down the shift key while dragging the mouse over a portion of the image will function as a zoom box]
8. Select the polygon or freehand tool, and then draw a training region in the reservoir. Keep in mind that you want your training region to capture any real diversity in the class but you want to avoid including any pixels that are outside of the class—e.g., peninsulas of land that jut into the reservoir. You do **not** need to perfectly select all of the pixels; the point is to get a representative sample.

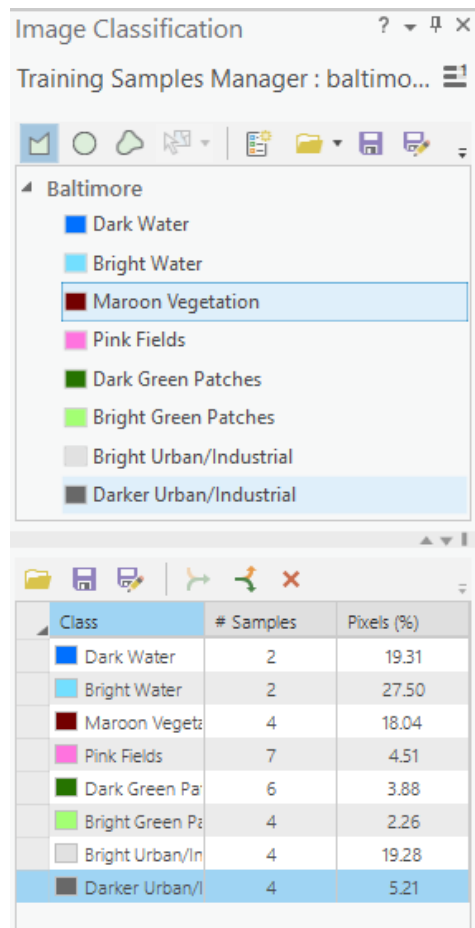


9. Once you double-click to close the polygon you will see it appear in your Training Sample Manager Dialog.
10. Zoom to another part of the image and find another “Dark Water” area—either another reservoir or parts of the Harbor that are particularly dark. Create a second polygon there.
 - Note: you can get out of drawing mode any time by hitting the esc key. That will make it easier for you to move around without accidentally creating training regions.
11. If you feel like you have not yet captured the full diversity of “Dark Water” in the image then create a third (or more) polygon in another water body. You can always remove polygons that you don’t like simply by highlighting them in the Training Sample Manager and deleting them.
12. Once you are satisfied with the number of polygons, select all rows in the Training Sample Manager (Shift + Click) and apply “Collapse” () to merge them into one training set for easy management.
13. Use “Save As” to save your Schema and your Training Regions to the location of your choice. You want to save often during this process!



The "save as" icon

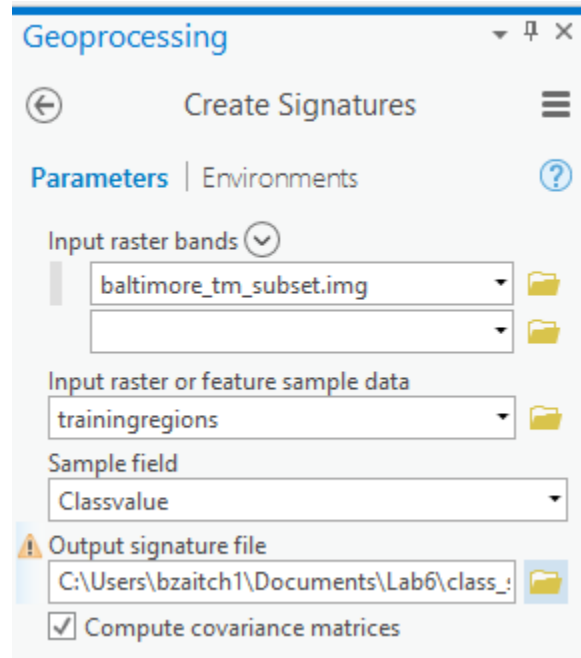
Now you'll need to repeat steps 5-12 for all other land cover classes, repeatedly adding new classes and giving each its own color and number. There's an art to this: you don't want "contamination" in your training regions, but you do want to capture the full range of variability for the class that each training region describes. There's no limit on how many separate polygons you can use to define a merged training region: for a large uniform feature you might get away with one, but for a fragmented land use class you might want to use 5 or more. **Save often**, overwriting your previous file each time. Note that you want to save both the "schema" (top Save option in the figure below) and the training regions (lower Save in the figure below). When you're done with all eight classes, the Training Sample Manager should look something like this:



Save your training regions. Breathe a sigh of relief.

Classify the Image

1. Activate the *Classify* tool under Imagery → Classification Tools.
2. Select “Maximum Likelihood” as your Classifier algorithm¹ (you might need to scroll up to see it), and load your training samples file². Specify output location and a file name that will remind you that this is your supervised classification, and then run the classification.
3. Create an ascii “signatures” file from your classes using Analysis → Tools → Toolboxes → Spatial Analyst Tools → Multivariate → Create Signatures. Inputs are as follows:



4. Open your signature file in Notepad. Notice that this file gives you information both on the mean spectral signature of each class and on variance/covariance of Landsat bands for all pixels in your training region. The latter provides a sense of how diverse each class is, and in cases where you aren't happy with your classification results it can be used as a guide for merging, expanding, or splitting training classes.
5. Generate a Dendrogram of your training classes using Analysis → Tools → Toolboxes → Spatial Analyst Tools → Multivariate → Dendrogram. Your input will be the signature file you just created. Your output is a text file: store it somewhere that you'll be able to find it. Accept other defaults.

¹ Last year, a seemingly random subset of students reported a bug in the Maximum Likelihood classification that we haven't been able to figure out. If your Maximum Likelihood classification returns everything in a single class then use the SVM classification algorithm as a back-up.

² To load the training samples file, you might need to go to Catalog (tab at the bottom of the window) and refresh the folder where it is saved by right clicking on the name of the folder and pressing “Refresh.” If you don't see your folder listed then you may need to add a folder connection by right clicking on “Folders” and “Add Folder Connection.”

6. Open the Dendrogram output. A Dendrogram is a plot that shows the merging structure of training classes: classes that are most different from each other split first (top of the tree) while more similar classes split last.

Q1: Copy your dendrogram onto your answer sheet (probably doing it as a picture rather than text will be easiest). Which two classes are most similar in your image? Does this make sense?

Q2: Pan around your classified image, zooming in on various features. How did you do? Are there any obvious errors (e.g., water bodies that have been classified as “pink fields”)? If there are systematic errors, can you explain them based on the reflectance properties of the classes in which errors occurred?

Q3: What are “Maroon Vegetation,” “Pink Fields,” “Dark Green Vegetation,” and “Bright Green Fields”?

Q4: This is an October image. Have the trees lost their leaves yet? How does this impact your ability to classify the image? Are there some land cover details that might be easier to detect in a different season? [You might want to add P15R3320100123.img, a winter image, to your Map Document when considering these questions]

Q5: Report the area of each class (including units!). This can be calculated using Toolboxes → Spatial Analyst Tools → Zonal → Zonal Statistics as Table, specifying Classvalue as your Zone field and your classification as the input raster and input value raster. The output table will appear under Contents, and you can right click to open it.

Q6: Compare your class areas with those of another student. If there are systematic differences between the classifications, can the two of you explain them on the basis of the training regions that you created for each class?

Unsupervised Classification

This one is easy: Under Classification Tools → Classify select ISO Cluster Classifier. Set the maximum class number to 8, set an output name that will identify this as your unsupervised classification, and accept all other defaults. If you have time you can compare this to Support Vector Machine and Random Tree classifiers.

Q7: Comment on the similarities and differences between your Supervised and Unsupervised Images. Is either one superior than the other for particular classes? [Note: you can dig into the details using Spatial Analyst → Zonal → Tabulate Areas, but that’s optional]

Q8: Include a screen capture of both your supervised and unsupervised classifications in the lab report.

Post-Processing a Classified Image

The results you have are good enough for our purposes, but keep in mind these handy tricks for later:

- If you want to combine classes (e.g., dark water and bright water) you can always do so using Classification Tools → Merge Classes

- Speckling can be removed using a Majority Filter, and rough edges can be smoothed using Boundary Clean. Both are found in Analysis → Tools → Toolboxes → Spatial Analyst Tools → Generalization. Another way to approach this problem is by segmenting your image prior to classification, an option that is available under the Classification Tools.
- Small isolated regions can be cleaned out of a classification using the Region Group tool under Toolboxes → Spatial Analyst Tools → Generalization.

Q9: Last week, Kevin performed a ground truth of his own classification of the Baltimore area. Using a Rover, he collected 584 ground control points and generated the Confusion Matrix below. Calculate the Overall Accuracy, Kappa Coefficient, as well as the User's and Producer's Accuracy for each class. Which class caused the most problems from a Producer's perspective (omission)? Which caused the most problems from a User's perspective (commission)?

Ground Truth

Satellite-derived Map		Water	Dense Urban	Mixed industrial	Coniferous Forest	Broadleaf Forest	Suburban Residential	Agriculture	Total
	Water	92	14	4	0	0	0	0	110
	Dense Urban	2	42	3	0	0	1	0	48
	Mixed Industrial	1	8	55	0	0	15	0	79
	Coniferous Forest	1	0	0	41	2	6	1	51
	Broadleaf Forest	0	0	2	11	75	9	2	99
	Suburban Residential	0	2	1	5	4	141	3	156
	Agriculture	0	0	0	0	1	4	36	41
	Total	96	66	65	57	82	176	42	584